Consolidated Cut blocks (November 2024)

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1. Description

The consolidated cutblocks dataset spatially combines currently reported or detected forest harvesting data into a single analysis ready spatial dataset. The following datasets are the inputs to the consolidated cutblocks dataset:

- Reporting Silviculture Updates and Land Status Tracking System (RESULTS)
- Landsat satellite change detection
- Forest Cover Inventory (VRI)

The RESULTS system contains the reported harvest and silviculture obligations from licensees on record going back to 1900. Licensees are required to report attribute information of all harvesting activity in the previous fiscal year to RESULTS by June 1st each year. Spatial harvest data is only required to be reported by June 1st after the year harvesting has been completed, although licensees may voluntarily submit spatial harvest information before harvest completion of an opening. As a result, openings with harvest activity occurring over multiple years may take multiple years after harvesting has started before spatial information appears in the RESULTS system. The consolidated cutblocks dataset includes only RESULTS cutblocks with spatial information in the system on the date the dataset is run (quarterly). This results in some multiyear cutblocks appearing in dataset years after harvest start year of the RESULTS submission.

RESULTS has cutblock records going back to 1900. It should also be noted that older records in the system may not be as complete as current harvest records. Furthermore, harvest on private lands and in certain licence types is not required to be reported to the RESULTS system.

Landsat satellite change detection datasets (created annually from 1999 to 2023 and quarterly since 2023) have been incorporated into the consolidated cutblocks dataset to identify harvest in those areas not reported in RESULTS (e.g. unreported harvest, harvest in private lands and harvest in licence types without reporting obligations). For cutblocks detected using this method, harvest start and completion dates are estimated using regularly monitored satellite imagery. For more information on the change detection process see https://www2.gov.bc.ca/gov/content/industry/forestry/managing-our-forest-resources/forest-inventory/forest-cover-inventories/forest-disturbance-monitoring .

Further harvesting is also detected in the Forest Cover Inventory (VRI) dataset through various inventory collection methods. These are mainly older cutblocks that do not have spatial records in RESULTS.

As mentioned above, older harvest is more likely to be missing from the input datasets as complete records and detection technologies were not always available. Older cutblocks in the data are likely to be less spatially accurate than newer data because of evolving collection technologies and updated data standards. Removing forest reserves from the cutblock data, for example, is more challenging in older records since the reporting standards on reserves were less consistent pre-2012. As a result, forest reserves are more likely to be included within the older polygons in the consolidated cutblocks dataset.

The methodology used to combine the three main data sources has been updated to provide more precise and current harvest information when available. A description of the updated methodology is in the Updated Methodology sections of this document.

2. Data Inputs

Name	Source
VRI	WHSE_FOREST_VEGETATION.VEG_COMP_LYR_R1_POLY (2024)
RESULTS Forest Cover	WHSE_FOREST_VEGETATION.RSLT_FOREST_COVER_INV_SVW
RESULTS Opening	**1.** WHSE_FOREST_VEGETATION.RSLT_OPENING_SVW **2.** WHSE_FOREST_VEGETATION.RSLT_OPENING_VW
RESULTS Activity Treatment	WHSE_FOREST_VEGETATION.RSLT_ACTIVITY_TREATMENT_SVW ACT
Landsat Change Detection	\\spatialfiles.bcgov\archive\for\VIC\HTS\FAIB_DATA_FOR_DISTRIBUTION\Cutblocks\Harvest_change_detection_files

3. Data Model (updated Nov, 2024)

Field Name	Data Type	Description
VEG_CONSOLIDATED_CUT_BLOCK_ID	Number	**VEG_CONSOLIDATED_CUT_BLOCK_ID** is a system generated unique identification number.
OPENING_ID	Integer	**OPENING_ID** is a system generated value assigned by RESULTS to uniquely identify the opening. When DATA_SOURCE = Satellite Imagery - Change Detection, the field value is NULL. When DATA_SOURCE = VRI, the field value is NULL unless OPENING_ID is reported within the Veg Comp Poly, in which case it contains the reported OPENING_ID.
HARVEST_START_DATE	Date	**HARVEST_START_DATE** represents the estimated start date of the logging activity of an opening. For polygons where DATA_SOURCE = RESULTS, the field is populated using the activity treatment start date (i.e. ATU_START_DATE) of the earliest logging disturbance of an opening. For polygons where DATA_SOURCE = Satellite Imagery - Change Detection, the field value is estimated from dates of the satellite images used in the change detection process. For polygons where DATA_SOURCE = VRI, the field

		is populated with the Veg Comp Poly Harvest_Year_Date field.
HARVEST_END_DATE	Date	**HARVEST_END_DATE** represents the estimated end date of logging activity of an opening. For polygons where DATA_SOURCE = RESULTS, the field is populated using the activity treatment completion date (i.e. ATU_COMPLETION_DATE) of the last logging disturbance of an opening. For polygons where DATA_SOURCE = Satellite Imagery - Change Detection, the field value is estimated from dates of the satellite images used in the change detection process. For polygons where DATA_SOURCE = VRI, the field is populated with the Veg Comp Poly Harvest_Year_Date field.
HARVEST_START_YEAR_CALENDAR	Integer	**HARV_START_YEAR_CALENDAR** is the calendar year of harvest start date of an opening. Derived from the HARVEST_START_DATE field.
HARVEST_MID_YEAR_CALENDAR	Integer	**HARV_MID_YEAR_CALENDAR** is the calendar year of the mid point between the harvest start date and harvest end date of an opening. Derived from the HARVEST_START_DATE and HARVEST_END_DATE fields.
HARVEST_START_YEAR_FISCAL	Integer	**HARVEST_START_YEAR_FISCAL** is the fiscal year of harvest start date of an opening. Derived from the HARVEST_START_DATE field.
HARVEST_MID_YEAR_FISCAL	Integer	**HARVEST_MID_YEAR_FISCAL** is the fiscal year of the mid point between the harvest start date and harvest end date of an opening. Derived from the HARVEST_START_DATE and HARVEST_END_DATE fields.
PERCENT_CLEARCUT	Integer	**PERCENT_CLEARCUT** is the estimated percent of the polygon where clearcut was used as the silvicultural system. When DATA_SOURCE = RESULTS, it is calculated by dividing the treatment area of activities with CLEAR (clearcut) or CCRES (clearcut with reserves) silviculture system codes by the total treatment area of logging activities. When DATA_SOURCE = VRI, it is assumed to be 100% unless OPENING_ID is

		present in the Veg Comp Poly, in which case the calculation method from RESULTS is used. When DATA_SOURCE = Satellite Imagery - Change Detection, the field value is assumed to be 100%.
PERCENT_PARTIAL_CUT	Integer	**PERCENT_PARTIAL_CUT** is the estimated percent of the opening where partial cut was used as the silvicultural system. When DATA_SOURCE = RESULTS, it is calculated by dividing the treatment area of activities with PATCH (patch cut), SELEC (selection), RETEN (retention), SEEDT (seed tree), or SHELT (shelterwood) silviculture system codes by the total treatment area of logging activities. When DATA_SOURCE = VRI, it is assumed to be 0%, unless OPENING_ID is present in the Veg Comp Poly, in which case the calculation method from RESULTS is used. When DATA_SOURCE = Satellite Imagery - Change Detection, it is assumed to be 0%.
DATA_SOURCE	Varchar (20)	**DATA_SOURCE** is the data source of the polygon and corresponding attribute information. The possible data sources are VRI, RESULTS, and Satellite Imagery - Change Detection.
DATA_SOURCE_DATE	Date	**DATA_SOURCE_DATE** is the date the data was pulled from the data source.
PUBLISHED_DATE	Date	**PUBLISHED_DATE** is this dataset was published
AREA_HA	Number	**AREA_HA** is the system calculated polygon area in Hectares, calculated to 2 decimal places.

4. Methodology: Summary of Nov 2024 Updates

The consolidated cutblocks methodology has been updated in 2024. The changes have been implemented to:

1. Increase the precision of the date of harvest
2. Add new fields to provide additional information on disturbance type, harvest year, and dataset currency (i.e. recency).
3. Provide the most current cutblock information sooner by incorporating change detection mapping into the dataset more frequently (quarterly).

The dataset is current to the data in the input datasets on that day.

5. Methodology: Detailed Description of Nov 2024 Updates

Increasing the precision of the date of harvest

The updated methodology increases the precision of the date of harvest using the following 3 methods:

1. Landsat satellite change detection harvest dates are used for RESULT openings that overlap with change detection polygons where the range of the RESULT disturbance start and end dates is greater than 4 months.

Why? – RESULTS opening disturbance dates used in the Consolidated Cutblocks dataset can span multiple years. Change detection detects smaller part of disturbances within openings resulting in more precise disturbance dates in most cases.

2. Spatial information from denudations in the RESULTS Activity Treatment dataset is used if available for openings where the range of the RESULTS opening disturbance start and end dates is greater than 12 months.

Why? – For some openings, licensees report spatial information of denudations for openings in parts. In these cases, denudations in the RESULTS Activity Treatment dataset have an activity treatment date that can be more precise than the generalized disturbance dates used for openings in consolidated cutblocks dataset.

Note: Licensees are required to report **attribute information** of denudations from the previous fiscal year by the June 1st reporting deadline. This information is tracked in the RESULTS Activity Treatment dataset. **Spatial information** of denudations can be submitted by licensees for the previous fiscal year, but it is not required. This updated methodology was only applied when spatial information of denudations is available in the RESULTS Activity Treatment dataset.

3. The method used to estimate harvest start and end dates from RESULT blocks has been updated

WHY? - The previous method for estimating harvest start and end dates included dates from a broader range of silviculture activities on an opening that may not have always been harvest. This resulted in a

larger harvest date range in some cases then in reality. The updated logic limits dates to data rows with a recorded silvicultural systems code that is used for the harvesting activity only. The updated logic is as follows:

```
Select
        min(ATU_START_DATE) AS  upd_disturbance_start_date,,
        max(ATU_COMPLETION_DATE) AS upd_disturbance_end_date,
FROM WHSE_FOREST_VEGETATION.RSLT_ACTIVITY_TREATMENT_UNIT TEMP
WHERE ((TEMP.SILV_BASE_CODE = 'DN' and TEMP.DISTURBANCE_CODE in ('L','R', 'S'))or
(TEMP.SILV_BASE_CODE = 'DN' and TEMP.SILV_SYSTEM_CODE is not null AND
TEMP.DISTURBANCE_CODE <> 'E' ))
GROUP BY OPENING_ID
```

4. **Figure 1 below details the logic tree for selecting the data source to use for the updated consolidated cutblocks dataset.**

Adding new fields to provide additional information on disturbance type, harvest year, and dataset currency.

New fields have been added to the dataset to provide additional information on disturbance type and dataset currency (i.e. recency), and to provide additional methods to calculate harvest year for reporting purposes (e.g. fiscal year). Additional fields include fields for the mid point of the disturbance start and disturbance end dates for both calendar and fiscal year and fields that indicate the estimated percentage of the opening where a partial cut system was used to harvest.

Table 1 includes the updated attribute field for the consolidated cutblocks dataset. New fields are highlighted in green and modified field names in orange. Note that the **HARVEST YEAR** field name was changed to **HARV_START_YEAR_CALENDAR** to distinguish it from the newly added fiscal year and mid-year fields.

Table 1. Updated data model for consolidated cutblocks dataset.

Existing Column Name	Data Type	Data Length
VEG_CONSOLIDATED_CUT_BLOCK_ID	Number	10
OPENING_ID	Integer	10
HARVEST_START_DATE	Date	7
HARVEST_END_DATE	Date	7
HARVEST_START_YEAR_CALENDAR	Integer	4
HARVEST_MID_YEAR_CALENDAR	Integer	4
HARVEST_START_YEAR_FISCAL	Integer	4
HARVEST_MID_YEAR_FISCAL	Integer	4
PERCENT_CLEARCUT	Integer	3
PERCENT_PARTIAL_CUT	Integer	3
DATA_SOURCE	Varchar2	20
DATA_SOURCE_DATE	Date	7
PUBLISHED_DATE	Date	7
AREA_HA	Number	10

6. Comparing Nov 2024 Methodology with Prior Methodology

The updated methodology uses different methods to report a more precise harvest than the previous versions of the dataset. As a result, a small shift of the previously reported area by harvest start year has occurred. Generally, this shift in area is no more than 1.5% per year. Table 2 below shows the difference of reported harvest area by year between the 2 methods when data was run on July 18th, 2024 for both methods from 2010 to 2024.

Table 3 quantifies the difference in harvest start year by method since 2018. Notably, in every year since 2018, much of the area has shifted to a harvest start year that is 1 or 2 years more recent in the new methodology.

Table 4 summarizes the harvest area in old seral areas compared by method.

Table 2. Annual harvest area difference by method 2010 to 2024

Harvest Start Year	Harvest Area (ha) - Old Method	Harvest Area (ha) -New Method	% Change
2010	196,701	198,607	1.0%
2011	223,786	224,358	0.3%
2012	218,526	218,912	0.2%
2013	234,750	233,498	-0.5%
2014	213,060	214,443	0.6%
2015	240,503	243,351	1.2%
2016	242,841	245,894	1.3%
2017	229,752	229,060	-0.3%
2018	232,560	233,426	0.4%
2019	170,393	171,040	0.4%
2020	188,652	189,286	0.3%
2021	182,660	186,365	2.0%
2022	151,028	151,967	0.6%
2023	97,636	99,099	1.5%
2024	18,576	18,714	0.7%

Table 3. Area of harvest start year change by method 2018 to 2024

Harvest start year – new method	New method harvest start year is 1 year more recent than old method	New method harvest start year is 2 years more recent than old method	New method harvest start year is 3 to 5 years more recent than old method	New method harvest start year is 5+ years more recent than old method	New method harvest start year is older than old method	New method harvest start year is same year as old method	Total area (ha) - new method
2024	38	14	105	89		18,468	18,714
2023	1,824	528	505	365	3	95,874	99,099
2022	920	705	804	694	1,049	147,795	151,967
2021	1,735	522	849	901	1,387	180,971	186,365
2020	1,203	1,378	399	741	1	185,564	189,286
2019	1,467	440	659	578	38	167,858	171,040
2018	2,470	794	84	1,378	62	228,638	233,426
2017	1,389	882	351	631	120	225,687	229,060
2016	1,938	794	2,121	945	105	239,991	245,894
2015	3,086	761	1,059	1,624	138	236,683	243,351
2014	2,601	1,241	1,344	612	99	208,546	214,443
2013	1,935	425	480	1,152	180	229,326	233,498
2012	1,937	800	693	908	36	214,538	218,912
2011	2,475	710	1,576	497	22	219,078	224,358
2010	2,162	1,788	1,133	539	23	192,962	198,607

Table 4. Area of harvest start year change by method from 2018 to 2024

Harvest Year	Old Seral – Old Method	Old Seral – New Method	% Diff
2015	65,635	66,982	2.1%
2016	55,763	57,449	3.0%
2017	53,386	53,190	-0.4%
2018	50,921	50,939	0.0%
2019	39,829	40,003	0.4%
2020	41,894	41,895	0.0%
2021	41,779	42,581	1.9%
2022	29,279	29,401	0.4%
2023	17,350	17,614	1.5%
2024	3,927	3,943	0.4%

7. Production Schedule for Change Detection Mapping

Change detection mapping will be run every quarter of the fiscal-year, allowing for future Consolidated Cutblocks datasets to be updated more frequently. The change detection process continues to improve providing more accurate representations of harvesting where information is missing from the RESULTS submissions.

8. Appendix A: SQL for Retrieving Inputs

RSLT_FOREST_COVER_INV_SVW

```
WITH actTreat1
AS (
        SELECT DISTINCT TEMP.OPENING_ID
                ,1 AS DENUDED
        FROM WHSE_FOREST_VEGETATION.RSLT_ACTIVITY_TREATMENT_UNIT TEMP
        LEFT JOIN WHSE_FOREST_VEGETATION.RSLT_OPENING_VW VIW2 ON TEMP.OPENING_ID =
VIW2.OPENING_ID
        WHERE (
                        TEMP.SILV_BASE_CODE = 'DN'
                        AND VIW2.FOREST_FILE_ID IS NOT NULL
                        AND VIW2.OPENING_CATEGORY_CODE NOT IN (
                                'NREQ'
                                ,'NDWL'
                                ,'NDAML'
                                ,'NDCF'
                                ,'NDFS'
                                ,'NDVML'
                                ,'SPEX'
                                ,'SMPCF'
                                ,'SMPEX'
                                ,'SMPFS'
                                ,'SMPML'
                                ,'SMPWL'
                                )
                        )
                OR (
                        TEMP.SILV_BASE_CODE = 'DN'
                        AND TEMP.DISTURBANCE_CODE IN (
                                'L'
                                ,'R'
                                ,'S'
                                )
                        )
                OR (
                        TEMP.SILV_BASE_CODE = 'DN'
                        AND TEMP.SILV_SYSTEM_CODE IS NOT NULL
                        AND TEMP.DISTURBANCE_CODE <> 'E'
                        )
        )
        ,NON_PRODUCTIVE
AS (
        SELECT DISTINCT FOREST_COVER_ID
                ,1 AS NON_PROD
        FROM WHSE_FOREST_VEGETATION.RSLT_FOREST_COVER_INV_SVW
        WHERE STOCKING_STATUS_CODE IN (
                        'A'
                        ,-- alpine
```

```
                    'AF'
                    ,-- alpine forest
                    'C'
                    ,-- cultivated
                    'G'
                    ,-- gravelbar
                    --            'IMM', -- immature
                    'L'
                    ,-- Lake
                    'M'
                    ,-- meadow
                    --            'MAT', -- mature
                    'NC'
                    ,-- non-commercial
                    'NF'
                    ,-- non-forest
                    --            'NP',  -- non-productive
                    --            'NSR', -- Not Satisfactorily Restocked
                    'OR'
                    ,-- Open range
                    'R'
                    ,-- rock
                    --            'RES', -- residual
                    'S'
                    ,-- swamp
                    'U' -- urban
                    )
             OR (
                    STOCKING_STATUS_CODE IN ('NP')
                    AND STOCKING_TYPE_CODE IN (
                           'NAT'
                           ,'FOR'
                           ,'BR'
                           )
                    )
       )
       ,RESERVES
AS (
       SELECT DISTINCT FOREST_COVER_ID
              ,1 AS RESERVE
       FROM WHSE_FOREST_VEGETATION.RSLT_FOREST_COVER_INV_SVW
       WHERE (
                    SILV_RESERVE_CODE IN ('G')
                    AND STOCKING_STATUS_CODE = 'MAT'
                    AND EXTRACT(YEAR FROM FOREST_COVER_WHEN_UPDATED) >= 2012
                    )
             OR (
                    SILV_RESERVE_CODE IN ('G')
                    AND STOCKING_STATUS_CODE IN (
                           'NP'
```

```
                        ,'MAT'
                        ,'IMM'
                        ,'RES'
                        )
                AND STOCKING_TYPE_CODE IN (
                        'NAT'
                        ,'FOR'
                        ,'BR'
                        )
                )
        OR (
                SILV_RESERVE_CODE IN (
                        'G'
                        ,'W'
                        )
                AND SILV_RESERVE_OBJECTIVE_CODE = 'WTR'
                AND STOCKING_STATUS_CODE IN (
                        'NP'
                        ,'MAT'
                        ,'IMM'
                        ,'RES'
                        )
                AND STOCKING_TYPE_CODE IN (
                        'NAT'
                        ,'FOR'
                        ,'BR'
                        )
                )
    )
    ,atu_dates
AS (
    SELECT OPENING_ID
        ,min(ATU_START_DATE) AS upd_disturbance_start_date
        ,max(ATU_COMPLETION_DATE) AS upd_disturbance_end_date
        ,sum(CASE
                WHEN SILV_SYSTEM_CODE IN (
                        'CLEAR'
                        ,'CCRES'
                        )
                    OR SILV_SYSTEM_CODE IS NULL
                    THEN TREATMENT_AMOUNT
                WHEN SILV_SYSTEM_CODE IS NULL
                    THEN NULL
                ELSE 0
                END) AS CLEARCUT_AREA
        ,round(sum(CASE
                    WHEN SILV_SYSTEM_CODE IN (
                            'CLEAR'
                            ,'CCRES'
                            )
```

```
                                        OR SILV_SYSTEM_CODE IS NULL
                                        THEN TREATMENT_AMOUNT
                                    ELSE 0
                                    END) / sum(TREATMENT_AMOUNT) * 100, 0) AS
CLEARCUT_AREA_perc
                ,sum(CASE
                            WHEN SILV_SYSTEM_CODE IN (
                                        'PATCT'
                                        ,'SELEC'
                                        ,'RETEN'
                                        ,'SEEDT'
                                        ,'SHELT'
                                        ,'COPPI'
                                        ,'IMCUT'
                                        )
                                    THEN TREATMENT_AMOUNT
                            ELSE 0
                            END) AS PARTIAL_CUT_AREA
                ,round(sum(CASE
                                    WHEN SILV_SYSTEM_CODE IN (
                                                'PATCT'
                                                ,'SELEC'
                                                ,'RETEN'
                                                ,'SEEDT'
                                                ,'SHELT'
                                                ,'COPPI'
                                                ,'IMCUT'
                                                )
                                        THEN TREATMENT_AMOUNT
                                    ELSE 0
                                    END) / sum(TREATMENT_AMOUNT) * 100, 0) AS
PARTIAL_CUT_AREA_PERC
                ,sum(TREATMENT_AMOUNT) AS total_treatment_area
        FROM WHSE_FOREST_VEGETATION.RSLT_ACTIVITY_TREATMENT_UNIT TEMP
        WHERE (
                        (
                            TEMP.SILV_BASE_CODE = 'DN'
                            AND TEMP.DISTURBANCE_CODE IN (
                                    'L'
                                    ,'R'
                                    ,'S'
                                    )
                            )
                        OR (
                            TEMP.SILV_BASE_CODE = 'DN'
                            AND TEMP.SILV_SYSTEM_CODE IS NOT NULL
                            AND TEMP.DISTURBANCE_CODE <> 'E'
                            )
                        )
                AND TREATMENT_AMOUNT > 0
```

```
	GROUP BY OPENING_ID
	)
	,upd_atu_dates
AS (
	SELECT opening_id
		,CASE
			WHEN upd_disturbance_start_date > upd_disturbance_end_date
				THEN upd_disturbance_end_date
			ELSE upd_disturbance_start_date
			END AS upd_disturbance_start_date
		,CASE
			WHEN upd_disturbance_start_date > upd_disturbance_end_date
				THEN upd_disturbance_start_date
			ELSE upd_disturbance_end_date
			END AS upd_disturbance_end_date
		,CLEARCUT_AREA
		,CLEARCUT_AREA_perc
		,PARTIAL_CUT_AREA
		,CASE
			WHEN CLEARCUT_AREA_perc + PARTIAL_CUT_AREA_PERC > 100
				THEN PARTIAL_CUT_AREA_PERC - (CLEARCUT_AREA_perc +
PARTIAL_CUT_AREA_PERC - 100)
			ELSE PARTIAL_CUT_AREA_PERC
			END AS PARTIAL_CUT_AREA_PERC
		,total_treatment_area
	FROM atu_dates
	)
	,calc_fiscal_dates
AS (
	SELECT CASE
			WHEN upd_disturbance_start_date IS NOT NULL
				AND (
					upd_disturbance_start_date <=
upd_disturbance_end_date
					OR upd_disturbance_end_date IS NULL
					)
				THEN upd_disturbance_start_date
			WHEN (upd_disturbance_start_date > upd_disturbance_end_date)
				OR (
					upd_disturbance_start_date IS NULL
					AND upd_disturbance_end_date IS NOT NULL
					)
				THEN upd_disturbance_end_date
			WHEN DISTURBANCE_START_DATE IS NOT NULL
				AND (
					DISTURBANCE_START_DATE <= DISTURBANCE_END_DATE
					OR DISTURBANCE_END_DATE IS NULL
					)
				THEN DISTURBANCE_START_DATE
			WHEN (DISTURBANCE_START_DATE > DISTURBANCE_END_DATE)
```

```
                        OR (
                                DISTURBANCE_START_DATE IS NULL
                                AND DISTURBANCE_END_DATE IS NOT NULL
                                )
                        THEN DISTURBANCE_END_DATE
                ELSE upd_disturbance_start_date
                END AS DISTURBANCE_START_DATE
        ,CASE
                WHEN upd_disturbance_end_date IS NOT NULL
                        AND (
                                upd_disturbance_start_date <=
upd_disturbance_end_date
                                OR upd_disturbance_start_date IS NULL
                                )
                        THEN upd_disturbance_end_date
                WHEN (upd_disturbance_start_date > upd_disturbance_end_date)
                        OR (
                                upd_disturbance_end_date IS NULL
                                AND upd_disturbance_start_date IS NOT NULL
                                )
                        THEN upd_disturbance_start_date
                WHEN DISTURBANCE_END_DATE IS NOT NULL
                        AND (
                                DISTURBANCE_START_DATE <= DISTURBANCE_END_DATE
                                OR DISTURBANCE_START_DATE IS NULL
                                )
                        THEN DISTURBANCE_END_DATE
                WHEN (DISTURBANCE_START_DATE > DISTURBANCE_END_DATE)
                        OR (
                                DISTURBANCE_END_DATE IS NULL
                                AND DISTURBANCE_START_DATE IS NOT NULL
                                )
                        THEN DISTURBANCE_START_DATE
                ELSE upd_disturbance_start_date
                END AS DISTURBANCE_END_DATE
        ,RSLT.DISTURBANCE_START_DATE AS DISTURBANCE_START_DATE_r
        ,RSLT.DISTURBANCE_END_DATE AS DISTURBANCE_END_DATE_r
        ,upd_disturbance_end_date
        ,upd_disturbance_start_date
        ,CASE
                WHEN CLEARCUT_AREA_perc IS NULL
                        THEN 100
                ELSE CLEARCUT_AREA_perc
                END AS CLEARCUT_AREA_perc
        ,CASE
                WHEN PARTIAL_CUT_AREA_PERC IS NULL
                        THEN 0
                ELSE PARTIAL_CUT_AREA_PERC
                END AS PARTIAL_CUT_AREA_PERC
        ,FC.GEOMETRY
```

```
              ,FC.OPENING_ID
              ,fc.forest_cover_id
              ,RSLT.DENUDATION_1_DISTURBANCE_CODE
              ,RSLT.DENUDATION_1_SILV_SYSTEM_CODE
              ,RSLT.DENUDATION_1_SILV_VARIANT_CODE
              ,RSLT.DENUDATION_1_CUT_PHASE_CODE
              ,RSLT.DENUDATION_2_DISTURBANCE_CODE
              ,RSLT.DENUDATION_2_SILV_SYSTEM_CODE
              ,RSLT.DENUDATION_2_SILV_VARIANT_CODE
              ,RSLT.DENUDATION_2_CUT_PHASE_CODE
       FROM WHSE_FOREST_VEGETATION.RSLT_FOREST_COVER_INV_SVW FC
       LEFT OUTER JOIN ACTTREAT1 act1 ON FC.OPENING_ID = act1.OPENING_ID
       LEFT OUTER JOIN RESERVES RS ON FC.FOREST_COVER_ID = RS.FOREST_COVER_ID
       LEFT OUTER JOIN NON_PRODUCTIVE NP ON FC.FOREST_COVER_ID =
NP.FOREST_COVER_ID
       JOIN WHSE_FOREST_VEGETATION.RSLT_OPENING_VW RSLT ON FC.OPENING_ID =
RSLT.OPENING_ID
       LEFT JOIN upd_atu_dates upd_dates ON FC.OPENING_ID = upd_dates.OPENING_ID
       WHERE FC.GEOMETRY IS NOT NULL
              AND DENUDED = 1
              AND Reserve IS NULL
              AND Non_Prod IS NULL
       )
SELECT DISTURBANCE_START_DATE
       ,DISTURBANCE_END_DATE
       ,DISTURBANCE_START_DATE AS DISTURBANCE_START_DATE_r
       ,DISTURBANCE_END_DATE AS DISTURBANCE_END_DATE_r
       ,upd_disturbance_end_date
       ,upd_disturbance_start_date
       ,CASE
              WHEN CLEARCUT_AREA_perc IS NULL
                     THEN 100
              ELSE CLEARCUT_AREA_perc
              END AS CLEARCUT_AREA_perc
       ,CASE
              WHEN PARTIAL_CUT_AREA_PERC IS NULL
                     THEN 0
              ELSE PARTIAL_CUT_AREA_PERC
              END AS PARTIAL_CUT_AREA_PERC
       ,GEOMETRY
       ,OPENING_ID
       ,forest_cover_id
       ,DISTURBANCE_START_DATE + (DISTURBANCE_END_DATE - DISTURBANCE_START_DATE) /
2 AS mid_date
       ,extract(YEAR FROM DISTURBANCE_START_DATE) AS HARVEST_YEAR_START
       ,extract(YEAR FROM DISTURBANCE_END_DATE) AS HARVEST_YEAR_END
       ,EXTRACT(YEAR FROM (DISTURBANCE_START_DATE + (DISTURBANCE_END_DATE -
DISTURBANCE_START_DATE) / 2)) AS HARVEST_YEAR_MID
       ,EXTRACT(YEAR FROM ADD_MONTHS(DISTURBANCE_START_DATE, - 3)) || '-' ||
EXTRACT(YEAR FROM ADD_MONTHS(DISTURBANCE_START_DATE, 9)) AS fiscal_year_start
```

```
        ,EXTRACT(YEAR FROM ADD_MONTHS(DISTURBANCE_END_DATE, - 3)) || '-' ||
EXTRACT(YEAR FROM ADD_MONTHS(DISTURBANCE_END_DATE, 9)) AS fiscal_year_end
        ,EXTRACT(YEAR FROM ADD_MONTHS((DISTURBANCE_START_DATE +
(DISTURBANCE_END_DATE - DISTURBANCE_START_DATE) / 2), - 3)) || '-' || EXTRACT(YEAR
FROM ADD_MONTHS((DISTURBANCE_START_DATE + (DISTURBANCE_END_DATE -
DISTURBANCE_START_DATE) / 2), 9)) AS fiscal_year_mid
        ,MONTHS_BETWEEN(DISTURBANCE_END_DATE, DISTURBANCE_START_DATE) AS
month_betw_dates
        ,DENUDATION_1_DISTURBANCE_CODE
        ,DENUDATION_1_SILV_SYSTEM_CODE
        ,DENUDATION_1_SILV_VARIANT_CODE
        ,DENUDATION_1_CUT_PHASE_CODE
        ,DENUDATION_2_DISTURBANCE_CODE
        ,DENUDATION_2_SILV_SYSTEM_CODE
        ,DENUDATION_2_SILV_VARIANT_CODE
        ,DENUDATION_2_CUT_PHASE_CODE
FROM calc_fiscal_dates
WHERE disturbance_start_date IS NOT NULL
```

RSLT_OPENING_SVW

```
WITH actTreat1
AS (
        SELECT DISTINCT TEMP.OPENING_ID
                ,1 AS DENUDED
        FROM WHSE_FOREST_VEGETATION.RSLT_ACTIVITY_TREATMENT_UNIT TEMP
        JOIN WHSE_FOREST_VEGETATION.RSLT_OPENING_VW VIW2 ON TEMP.OPENING_ID =
VIW2.OPENING_ID
        WHERE (
                        TEMP.SILV_BASE_CODE = 'DN'
                        AND VIW2.FOREST_FILE_ID IS NOT NULL
                        AND VIW2.OPENING_CATEGORY_CODE NOT IN (
                                'NREQ'
                                ,'NDWL'
                                ,'NDAML'
                                ,'NDCF'
                                ,'NDFS'
                                ,'NDVML'
                                ,'SPEX'
                                ,'SMPCF'
                                ,'SMPEX'
                                ,'SMPFS'
                                ,'SMPML'
                                ,'SMPWL'
                                )
                        )
                OR (
                        TEMP.SILV_BASE_CODE = 'DN'
                        AND TEMP.DISTURBANCE_CODE IN (
```

```
                        'L'
                        ,'R'
                        ,'S'
                        )
                )
            OR (
                TEMP.SILV_BASE_CODE = 'DN'
                AND TEMP.SILV_SYSTEM_CODE IS NOT NULL
                AND TEMP.DISTURBANCE_CODE <> 'E'
                )
        )
        ,actDate
AS (
        SELECT a.OPENING_ID
            ,a.ACTUAL_TREATMENT_AREA
            ,a.ATU_START_DATE AS ATU_START_DATE_MAJORITY
            ,a.ATU_COMPLETION_DATE AS ATU_COMPLETION_DATE_MAJORITY
            ,ranked_order
        FROM (
            SELECT OPENING_ID
                ,ACTUAL_TREATMENT_AREA
                ,ATU_START_DATE
                ,ATU_COMPLETION_DATE
                ,ROW_NUMBER() OVER (
                    PARTITION BY OPENING_ID ORDER BY ACTUAL_TREATMENT_AREA DESC
                    ) ranked_order
            FROM (
                SELECT OPENING_ID
                    ,sum(coalesce(ACTUAL_TREATMENT_AREA, 0)) AS
ACTUAL_TREATMENT_AREA
                    ,ATU_START_DATE
                    ,ATU_COMPLETION_DATE
                FROM WHSE_FOREST_VEGETATION.RSLT_ACTIVITY_TREATMENT_SVW TEMP
                WHERE (
                        TEMP.SILV_BASE_CODE = 'DN'
                        AND TEMP.DISTURBANCE_CODE IN (
                            'L'
                            ,'R'
                            ,'S'
                            )
                        )
                    OR (
                        TEMP.SILV_BASE_CODE = 'DN'
                        AND TEMP.SILV_SYSTEM_CODE IS NOT NULL
                        AND TEMP.DISTURBANCE_CODE <> 'E'
                        )
                GROUP BY OPENING_ID
                    ,ATU_START_DATE
                    ,ATU_COMPLETION_DATE
                ) b
```

```
                ) a
        WHERE a.ranked_order = 1
        )
        ,fcSel
AS (
        SELECT DISTINCT OPENING_ID
                ,1 AS RFC
        FROM WHSE_FOREST_VEGETATION.RSLT_FOREST_COVER_INV_SVW
        WHERE GEOMETRY IS NOT NULL
        )
        ,atu_dates
AS (
        SELECT OPENING_ID
                ,min(ATU_START_DATE) AS upd_disturbance_start_date
                ,max(ATU_COMPLETION_DATE) AS upd_disturbance_end_date
                ,sum(CASE
                        WHEN SILV_SYSTEM_CODE IN (
                                'CLEAR'
                                ,'CCRES'
                                )
                                OR SILV_SYSTEM_CODE IS NULL
                                THEN TREATMENT_AMOUNT
                        WHEN SILV_SYSTEM_CODE IS NULL
                                THEN NULL
                        ELSE 0
                        END) AS CLEARCUT_AREA
                ,round(sum(CASE
                                WHEN SILV_SYSTEM_CODE IN (
                                        'CLEAR'
                                        ,'CCRES'
                                        )
                                        OR SILV_SYSTEM_CODE IS NULL
                                        THEN TREATMENT_AMOUNT
                                ELSE 0
                                END) / sum(TREATMENT_AMOUNT) * 100, 0) AS
CLEARCUT_AREA_perc
                ,sum(CASE
                        WHEN SILV_SYSTEM_CODE IN (
                                'PATCT'
                                ,'SELEC'
                                ,'RETEN'
                                ,'SEEDT'
                                ,'SHELT'
                                ,'COPPI'
                                ,'IMCUT'
                                )
                                THEN TREATMENT_AMOUNT
                        ELSE 0
                        END) AS PARTIAL_CUT_AREA
                ,round(sum(CASE
```

```
                                WHEN SILV_SYSTEM_CODE IN (
                                              'PATCT'
                                              ,'SELEC'
                                              ,'RETEN'
                                              ,'SEEDT'
                                              ,'SHELT'
                                              ,'COPPI'
                                              ,'IMCUT'
                                              )
                                       THEN TREATMENT_AMOUNT
                                ELSE 0
                                END) / sum(TREATMENT_AMOUNT) * 100, 0) AS
PARTIAL_CUT_AREA_PERC
              ,sum(TREATMENT_AMOUNT) AS total_treatment_area
       FROM WHSE_FOREST_VEGETATION.RSLT_ACTIVITY_TREATMENT_UNIT TEMP
       WHERE (
                     (
                            TEMP.SILV_BASE_CODE = 'DN'
                            AND TEMP.DISTURBANCE_CODE IN (
                                   'L'
                                   ,'R'
                                   ,'S'
                                   )
                            )
                     OR (
                            TEMP.SILV_BASE_CODE = 'DN'
                            AND TEMP.SILV_SYSTEM_CODE IS NOT NULL
                            AND TEMP.DISTURBANCE_CODE <> 'E'
                            )
                     )
              AND TREATMENT_AMOUNT > 0
       GROUP BY OPENING_ID
       )
       ,upd_dates
AS (
       SELECT opening_id
              ,CASE
                     WHEN upd_disturbance_start_date > upd_disturbance_end_date
                            THEN upd_disturbance_end_date
                     ELSE upd_disturbance_start_date
                     END AS upd_disturbance_start_date
              ,CASE
                     WHEN upd_disturbance_start_date > upd_disturbance_end_date
                            THEN upd_disturbance_start_date
                     ELSE upd_disturbance_end_date
                     END AS upd_disturbance_end_date
              ,CLEARCUT_AREA
              ,CLEARCUT_AREA_perc
              ,PARTIAL_CUT_AREA
              ,CASE
```

```
                    WHEN CLEARCUT_AREA_perc + PARTIAL_CUT_AREA_PERC > 100
                           THEN PARTIAL_CUT_AREA_PERC - (CLEARCUT_AREA_perc +
PARTIAL_CUT_AREA_PERC - 100)
                    ELSE PARTIAL_CUT_AREA_PERC
                    END AS PARTIAL_CUT_AREA_PERC
             ,total_treatment_area
       FROM atu_dates
       )
       ,calc_fiscal_dates
AS (
       SELECT RSLT.OPENING_ID
             ,RSLT.GEOMETRY
             ,CASE
                    WHEN upd_disturbance_start_date IS NOT NULL
                           AND (
                                  upd_disturbance_start_date <= upd_disturbance_end_date
                                  OR upd_disturbance_end_date IS NULL
                                  )
                           THEN upd_disturbance_start_date
                    WHEN (upd_disturbance_start_date > upd_disturbance_end_date)
                           OR (
                                  upd_disturbance_start_date IS NULL
                                  AND upd_disturbance_end_date IS NOT NULL
                                  )
                           THEN upd_disturbance_end_date
                    WHEN DISTURBANCE_START_DATE IS NOT NULL
                           AND (
                                  DISTURBANCE_START_DATE <= DISTURBANCE_END_DATE
                                  OR DISTURBANCE_END_DATE IS NULL
                                  )
                           THEN DISTURBANCE_START_DATE
                    WHEN (DISTURBANCE_START_DATE > DISTURBANCE_END_DATE)
                           OR (
                                  DISTURBANCE_START_DATE IS NULL
                                  AND DISTURBANCE_END_DATE IS NOT NULL
                                  )
                           THEN DISTURBANCE_END_DATE
                    ELSE upd_disturbance_start_date
                    END AS DISTURBANCE_START_DATE
             ,CASE
                    WHEN upd_disturbance_end_date IS NOT NULL
                           AND (
                                  upd_disturbance_start_date <= upd_disturbance_end_date
                                  OR upd_disturbance_start_date IS NULL
                                  )
                           THEN upd_disturbance_end_date
                    WHEN (upd_disturbance_start_date > upd_disturbance_end_date)
                           OR (
                                  upd_disturbance_end_date IS NULL
                                  AND upd_disturbance_start_date IS NOT NULL
```

```
                                )
                        THEN upd_disturbance_start_date
                WHEN DISTURBANCE_END_DATE IS NOT NULL
                        AND (
                                DISTURBANCE_START_DATE <= DISTURBANCE_END_DATE
                                OR DISTURBANCE_START_DATE IS NULL
                                )
                        THEN DISTURBANCE_END_DATE
                WHEN (DISTURBANCE_START_DATE > DISTURBANCE_END_DATE)
                        OR (
                                DISTURBANCE_END_DATE IS NULL
                                AND DISTURBANCE_START_DATE IS NOT NULL
                                )
                        THEN DISTURBANCE_START_DATE
                ELSE upd_disturbance_start_date
                END AS DISTURBANCE_END_DATE
        ,CASE
                WHEN CLEARCUT_AREA_perc IS NULL
                        THEN 100
                ELSE CLEARCUT_AREA_perc
                END AS CLEARCUT_AREA_perc
        ,CASE
                WHEN PARTIAL_CUT_AREA_PERC IS NULL
                        THEN 0
                ELSE PARTIAL_CUT_AREA_PERC
                END AS PARTIAL_CUT_AREA_PERC
        ,RSLT.DENUDATION_1_DISTURBANCE_CODE
        ,RSLT.DENUDATION_1_SILV_SYSTEM_CODE
        ,RSLT.DENUDATION_1_SILV_VARIANT_CODE
        ,RSLT.DENUDATION_1_CUT_PHASE_CODE
        ,RSLT.DENUDATION_2_DISTURBANCE_CODE
        ,RSLT.DENUDATION_2_SILV_SYSTEM_CODE
        ,RSLT.DENUDATION_2_SILV_VARIANT_CODE
        ,RSLT.DENUDATION_2_CUT_PHASE_CODE
    FROM WHSE_FOREST_VEGETATION.RSLT_OPENING_SVW RSLT
    LEFT JOIN actTreat1 act ON RSLT.OPENING_ID = ACT.OPENING_ID
    LEFT JOIN FCSEL fc ON RSLT.OPENING_ID = FC.OPENING_ID
    LEFT JOIN actDate actd ON RSLT.OPENING_ID = actd.OPENING_ID
    LEFT JOIN upd_dates ON RSLT.OPENING_ID = upd_dates.OPENING_ID
    WHERE RSLT.GEOMETRY IS NOT NULL
        AND DENUDED = 1
        AND RFC IS NULL
    )
SELECT OPENING_ID
    ,GEOMETRY
    ,DISTURBANCE_START_DATE
    ,DISTURBANCE_END_DATE
    ,CLEARCUT_AREA_perc
    ,PARTIAL_CUT_AREA_PERC
    ,DISTURBANCE_START_DATE + (DISTURBANCE_END_DATE - DISTURBANCE_START_DATE) / 2 AS
```

```
mid_date
       ,extract(YEAR FROM DISTURBANCE_START_DATE) AS HARVEST_YEAR_START
       ,extract(YEAR FROM DISTURBANCE_END_DATE) AS HARVEST_YEAR_END
       ,EXTRACT(YEAR FROM (DISTURBANCE_START_DATE + (DISTURBANCE_END_DATE -
DISTURBANCE_START_DATE) / 2)) AS HARVEST_YEAR_MID
       ,EXTRACT(YEAR FROM ADD_MONTHS(DISTURBANCE_START_DATE, - 3)) || '-' || EXTRACT(YEAR
FROM ADD_MONTHS(DISTURBANCE_START_DATE, 9)) AS fiscal_year_start
       ,EXTRACT(YEAR FROM ADD_MONTHS(DISTURBANCE_END_DATE, - 3)) || '-' || EXTRACT(YEAR
FROM ADD_MONTHS(DISTURBANCE_END_DATE, 9)) AS fiscal_year_end
       ,EXTRACT(YEAR FROM ADD_MONTHS((DISTURBANCE_START_DATE + (DISTURBANCE_END_DATE -
DISTURBANCE_START_DATE) / 2), - 3)) || '-' || EXTRACT(YEAR FROM
ADD_MONTHS((DISTURBANCE_START_DATE + (DISTURBANCE_END_DATE - DISTURBANCE_START_DATE) /
2), 9)) AS fiscal_year_mid
       ,MONTHS_BETWEEN(DISTURBANCE_END_DATE, DISTURBANCE_START_DATE) AS month_betw_dates
       ,DENUDATION_1_DISTURBANCE_CODE
       ,DENUDATION_1_SILV_SYSTEM_CODE
       ,DENUDATION_1_SILV_VARIANT_CODE
       ,DENUDATION_1_CUT_PHASE_CODE
       ,DENUDATION_2_DISTURBANCE_CODE
       ,DENUDATION_2_SILV_SYSTEM_CODE
       ,DENUDATION_2_SILV_VARIANT_CODE
       ,DENUDATION_2_CUT_PHASE_CODE
FROM calc_fiscal_dates
WHERE DISTURBANCE_START_DATE IS NOT NULL
```

```
RSLT_ACTIVITY_TREATMENT_SVW
WITH t1
AS (
       SELECT TEMP.GEOMETRY
              ,TEMP.OPENING_ID
              ,CASE
                     WHEN SILV_SYSTEM_CODE IN (
                                   'CLEAR'
                                   ,'CCRES'
                                   )
                            OR SILV_SYSTEM_CODE IS NULL
                            THEN 100
                     ELSE 0
                     END AS CLEARCUT_AREA_perc
              ,CASE
                     WHEN SILV_SYSTEM_CODE IN (
                                   'PATCT'
                                   ,'SELEC'
                                   ,'RETEN'
                                   ,'SEEDT'
                                   ,'SHELT'
                                   ,'COPPI'
```

```
                                ,'IMCUT'
                                )
                        THEN 100
                ELSE 0
                END AS PARTIAL_CUT_AREA_PERC
        ,FEATURE_AREA_SQM
        ,CASE
                WHEN ATU_START_DATE IS NOT NULL
                        AND (
                                ATU_START_DATE <= ATU_COMPLETION_DATE
                                OR ATU_COMPLETION_DATE IS NULL
                                )
                        THEN ATU_START_DATE
                WHEN (ATU_START_DATE > ATU_COMPLETION_DATE)
                        OR (
                                ATU_START_DATE IS NULL
                                AND ATU_COMPLETION_DATE IS NOT NULL
                                )
                        THEN ATU_COMPLETION_DATE
                ELSE ATU_START_DATE
                END AS ATU_START_DATE
        ,CASE
                WHEN ATU_COMPLETION_DATE IS NOT NULL
                        AND (
                                ATU_START_DATE <= ATU_COMPLETION_DATE
                                OR ATU_START_DATE IS NULL
                                )
                        THEN ATU_COMPLETION_DATE
                WHEN (ATU_START_DATE > ATU_COMPLETION_DATE)
                        OR (
                                ATU_COMPLETION_DATE IS NULL
                                AND ATU_START_DATE IS NOT NULL
                                )
                        THEN ATU_START_DATE
                ELSE ATU_COMPLETION_DATE
                END AS ATU_COMPLETION_DATE
        ,TEMP.DISTURBANCE_CODE
        ,TEMP.SILV_SYSTEM_CODE
        ,TEMP.SILV_SYSTEM_VARIANT_CODE
        ,TEMP.SILV_CUT_PHASE_CODE
        ,TEMP.SILV_BASE_CODE
FROM WHSE_FOREST_VEGETATION.RSLT_ACTIVITY_TREATMENT_SVW TEMP
WHERE extract(year FROM TEMP.ATU_START_DATE) >= 2012
        AND TEMP.GEOMETRY IS NOT NULL
        AND (
                (
                        TEMP.SILV_BASE_CODE = 'DN'
                        AND TEMP.DISTURBANCE_CODE IN (
                                'L'
                                ,'R'
```

```
                                ,'S'
                                )
                        )
                    OR (
                            TEMP.SILV_BASE_CODE = 'DN'
                            AND TEMP.SILV_SYSTEM_CODE IS NOT NULL
                            AND TEMP.DISTURBANCE_CODE <> 'E'
                            )
                    )
        )
SELECT GEOMETRY
        ,OPENING_ID
        ,ATU_START_DATE
        ,ATU_COMPLETION_DATE
        ,ATU_START_DATE + (ATU_COMPLETION_DATE - ATU_START_DATE) / 2 AS mid_date
        ,extract(YEAR FROM ATU_START_DATE) AS HARVEST_YEAR_START
        ,extract(YEAR FROM ATU_COMPLETION_DATE) AS HARVEST_YEAR_END
        ,EXTRACT(YEAR FROM (ATU_START_DATE + (ATU_COMPLETION_DATE - ATU_START_DATE) / 2))
AS HARVEST_YEAR_MID
        ,EXTRACT(YEAR FROM ADD_MONTHS(ATU_START_DATE, - 3)) || '-' || EXTRACT(YEAR FROM
ADD_MONTHS(ATU_START_DATE, 9)) AS fiscal_year_start
        ,EXTRACT(YEAR FROM ADD_MONTHS(ATU_COMPLETION_DATE, - 3)) || '-' || EXTRACT(YEAR
FROM ADD_MONTHS(ATU_COMPLETION_DATE, 9)) AS fiscal_year_end
        ,EXTRACT(YEAR FROM ADD_MONTHS((ATU_START_DATE + (ATU_COMPLETION_DATE -
ATU_START_DATE) / 2), - 3)) || '-' || EXTRACT(YEAR FROM ADD_MONTHS((ATU_START_DATE +
(ATU_COMPLETION_DATE - ATU_START_DATE) / 2), 9)) AS fiscal_year_mid
        ,MONTHS_BETWEEN(ATU_COMPLETION_DATE, ATU_START_DATE) AS month_betw_dates
        ,disturbance_code
        ,SILV_SYSTEM_CODE
        ,SILV_SYSTEM_VARIANT_CODE
        ,SILV_CUT_PHASE_CODE
        ,SILV_BASE_CODE
        ,CLEARCUT_AREA_perc
        ,PARTIAL_CUT_AREA_PERC
FROM t1
```

Reserves and non-productive land from in RESULTS (Removed from analysis using GIS spatial overlays)

```
SELECT GEOMETRY
          ,OPENING_ID
          ,1 AS NON_PROD_OR_RESERVE
FROM WHSE_FOREST_VEGETATION.RSLT_FOREST_COVER_INV_SVW
WHERE STOCKING_STATUS_CODE IN (
                  'A'
                  ,-- alpine
                  'AF'
                  ,-- alpine forest
                  'C'
                  ,-- cultivated
                  'G'
                  ,-- gravelbar
                  --                'IMM',   -- immature
                  'L'
                  ,-- Lake
```

```
                'M'
                ,-- meadow
                --                'MAT',   -- mature
                'NC'
                ,-- non-commercial
                'NF'
                ,-- non-forest
                --                'NP',    -- non-productive
                --                'NSR',   -- Not Satisfactorily Restocked
                'OR'
                ,-- Open range
                'R'
                ,-- rock
                --                'RES',   -- residual
                'S'
                ,-- swamp
                'U' -- urban
                )
        OR (
                STOCKING_STATUS_CODE IN ('NP')
                AND STOCKING_TYPE_CODE IN (
                        'NAT'
                        ,'FOR'
                        ,'BR'
                        )
                )
        OR (
                SILV_RESERVE_CODE IN ('G')
                AND STOCKING_STATUS_CODE = 'MAT'
                AND EXTRACT(YEAR FROM FOREST_COVER_WHEN_UPDATED) >= 2012
                )
        OR (
                SILV_RESERVE_CODE IN ('G')
                AND STOCKING_STATUS_CODE IN (
                        'NP'
                        ,'MAT'
                        ,'IMM'
                        ,'RES'
                        )
                AND STOCKING_TYPE_CODE IN (
                        'NAT'
                        ,'FOR'
                        ,'BR'
                        )
                )
        OR (
                SILV_RESERVE_CODE IN (
                        'G'
                        ,'W'
                        )
                AND SILV_RESERVE_OBJECTIVE_CODE = 'WTR'
                AND STOCKING_STATUS_CODE IN (
                        'NP'
                        ,'MAT'
                        ,'IMM'
                        ,'RES'
                        )
                AND STOCKING_TYPE_CODE IN (
```

```
                        'NAT'
                        ,'FOR'
                        ,'BR'
                        )
                )
```

VRI

```
WITH actTreat1
AS (
        SELECT DISTINCT TEMP.OPENING_ID
                ,1 AS DENUDED
        FROM WHSE_FOREST_VEGETATION.RSLT_ACTIVITY_TREATMENT_UNIT TEMP
        JOIN WHSE_FOREST_VEGETATION.RSLT_OPENING_VW VIW2 ON TEMP.OPENING_ID =
VIW2.OPENING_ID
        WHERE (
                        TEMP.SILV_BASE_CODE = 'DN'
                        AND VIW2.FOREST_FILE_ID IS NOT NULL
                        AND VIW2.OPENING_CATEGORY_CODE NOT IN (
                                'NREQ'
                                ,'NDWL'
                                ,'NDAML'
                                ,'NDCF'
                                ,'NDFS'
                                ,'NDVML'
                                ,'SPEX'
                                ,'SMPCF'
                                ,'SMPEX'
                                ,'SMPFS'
                                ,'SMPML'
                                ,'SMPWL'
                                )
                        )
                OR (
                        TEMP.SILV_BASE_CODE = 'DN'
                        AND TEMP.DISTURBANCE_CODE IN (
                                'L'
                                ,'R'
                                ,'S'
                                )
                        )
                OR (
                        TEMP.SILV_BASE_CODE = 'DN'
                        AND TEMP.SILV_SYSTEM_CODE IS NOT NULL
                        AND TEMP.DISTURBANCE_CODE <> 'E'
                        )
        )
        ,fcSel
AS (
        SELECT DISTINCT fc.OPENING_ID
                ,1 AS RFC
        FROM WHSE_FOREST_VEGETATION.RSLT_FOREST_COVER_INV_SVW fc
        JOIN actTreat1 act ON fc.OPENING_ID = act.OPENING_ID
        WHERE GEOMETRY IS NOT NULL
        )
```

```
        ,RESSel
AS (
        SELECT DISTINCT RESVW.OPENING_ID
                ,1 AS RESVW
        FROM WHSE_FOREST_VEGETATION.RSLT_OPENING_SVW RESVW
        JOIN actTreat1 act ON RESVW.OPENING_ID = act.OPENING_ID
        WHERE GEOMETRY IS NOT NULL
        )
        ,atu_dates
AS (
        SELECT OPENING_ID
                ,min(ATU_START_DATE) AS upd_disturbance_start_date
                ,max(ATU_COMPLETION_DATE) AS upd_disturbance_end_date
                ,sum(CASE
                                WHEN SILV_SYSTEM_CODE IN (
                                                'CLEAR'
                                                ,'CCRES'
                                                )
                                        OR SILV_SYSTEM_CODE IS NULL
                                        THEN TREATMENT_AMOUNT
                                WHEN SILV_SYSTEM_CODE IS NULL
                                        THEN NULL
                                ELSE 0
                                END) AS CLEARCUT_AREA
                ,round(sum(CASE
                                        WHEN SILV_SYSTEM_CODE IN (
                                                        'CLEAR'
                                                        ,'CCRES'
                                                        )
                                                OR SILV_SYSTEM_CODE IS NULL
                                                THEN TREATMENT_AMOUNT
                                        ELSE 0
                                        END) / sum(TREATMENT_AMOUNT) * 100, 0)
AS CLEARCUT_AREA_perc
                ,sum(CASE
                                WHEN SILV_SYSTEM_CODE IN (
                                                'PATCT'
                                                ,'SELEC'
                                                ,'RETEN'
                                                ,'SEEDT'
                                                ,'SHELT'
                                                ,'COPPI'
                                                ,'IMCUT'
                                                )
                                        THEN TREATMENT_AMOUNT
                                ELSE 0
                                END) AS PARTIAL_CUT_AREA
                ,round(sum(CASE
                                        WHEN SILV_SYSTEM_CODE IN (
                                                        'PATCT'
                                                        ,'SELEC'
                                                        ,'RETEN'
                                                        ,'SEEDT'
                                                        ,'SHELT'
                                                        ,'COPPI'
                                                        ,'IMCUT'
                                                        )
```

```
                                                THEN TREATMENT_AMOUNT
                                        ELSE 0
                                        END) / sum(TREATMENT_AMOUNT) * 100, 0)
AS PARTIAL_CUT_AREA_PERC
                ,sum(TREATMENT_AMOUNT) AS total_treatment_area
        FROM WHSE_FOREST_VEGETATION.RSLT_ACTIVITY_TREATMENT_UNIT TEMP
        WHERE (
                        (
                                TEMP.SILV_BASE_CODE = 'DN'
                                AND TEMP.DISTURBANCE_CODE IN (
                                        'L'
                                        ,'R'
                                        ,'S'
                                        )
                                )
                        OR (
                                TEMP.SILV_BASE_CODE = 'DN'
                                AND TEMP.SILV_SYSTEM_CODE IS NOT NULL
                                AND TEMP.DISTURBANCE_CODE <> 'E'
                                )
                        )
                AND TREATMENT_AMOUNT > 0
        GROUP BY OPENING_ID
        )
        ,upd_dates
AS (
        SELECT opening_id
                ,CASE
                        WHEN upd_disturbance_start_date >
upd_disturbance_end_date
                                THEN upd_disturbance_end_date
                        ELSE upd_disturbance_start_date
                        END AS upd_disturbance_start_date
                ,CASE
                        WHEN upd_disturbance_start_date >
upd_disturbance_end_date
                                THEN upd_disturbance_start_date
                        ELSE upd_disturbance_end_date
                        END AS upd_disturbance_end_date
                ,CLEARCUT_AREA
                ,CLEARCUT_AREA_perc
                ,PARTIAL_CUT_AREA
                ,CASE
                        WHEN CLEARCUT_AREA_perc + PARTIAL_CUT_AREA_PERC > 100
                                THEN PARTIAL_CUT_AREA_PERC -
(CLEARCUT_AREA_perc + PARTIAL_CUT_AREA_PERC - 100)
                        ELSE PARTIAL_CUT_AREA_PERC
                        END AS PARTIAL_CUT_AREA_PERC
                ,total_treatment_area
        FROM atu_dates
        )
SELECT VRI.HARVEST_DATE
        ,VRI.OPENING_ID
        ,VRI.GEOMETRY
        ,CASE
                WHEN HARVEST_DATE IS NOT NULL
```

```
                        THEN EXTRACT(YEAR FROM ADD_MONTHS(HARVEST_DATE, - 3))
|| '-' || EXTRACT(YEAR FROM ADD_MONTHS(HARVEST_DATE, 9))
                ELSE NULL
                END AS fiscal_year
        ,CASE
                WHEN CLEARCUT_AREA_perc IS NULL
                        THEN 100
                ELSE CLEARCUT_AREA_perc
                END AS CLEARCUT_AREA_perc
        ,CASE
                WHEN PARTIAL_CUT_AREA_PERC IS NULL
                        THEN 0
                ELSE PARTIAL_CUT_AREA_PERC
                END AS PARTIAL_CUT_AREA_PERC
FROM WHSE_FOREST_VEGETATION.VEG_COMP_POLY VRI
LEFT OUTER JOIN FCSEL fc ON vri.OPENING_ID = FC.OPENING_ID
LEFT OUTER JOIN RESSel RSL ON vri.OPENING_ID = RSL.OPENING_ID
LEFT JOIN upd_dates ON VRI.OPENING_ID = upd_dates.OPENING_ID
WHERE VRI.HARVEST_DATE IS NOT NULL
        AND (TO_CHAR(VRI.HARVEST_DATE, 'YYYY') >= '1965')
        AND RFC IS NULL
        AND RESVW IS NULL
```